

## 27 Group 2

### A · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

For Group 2 compounds, ionic size and charge density explain solubility and thermal stability trends; hydration and lattice enthalpies compete in dissolving salts. Group 2 hydroxides become more soluble down the group; sulfate solubility decreases. Industrial/analytical uses depend on insolubility and acid-base behaviour.

#### Worked example: follow the reasoning

A barium chloride test detects sulfate through insoluble  $\text{BaSO}_4$ ; acidify appropriately to avoid carbonate interference.

#### Quick check · multiple choice

A Which compound forms a white insoluble precipitate in the sulfate test? [1]

A  $\text{Mg}(\text{OH})_2$

B  $\text{CaO}$

C  $\text{BaSO}_4$

D  $\text{Ba}(\text{OH})_2$

Write your choice: \_\_\_\_\_

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Predict solubility trend of  $\text{Mg}(\text{OH})_2$  to  $\text{Ba}(\text{OH})_2$  and explain why sulfate tests use barium salts. [3]

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## Answers and explanation

Quick check: C.  $\text{BaSO}_4$

### Question 1 [3 marks]

Hydroxide solubility increases down Group 2 [1];  $\text{BaSO}_4$  has very low solubility [1], forming a visible white precipitate with sulfate ions [1].

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.